Brigham and Women's Hospital
Founding Member, Mass General Brigham

Psychiatry Neuroimaging Laboratory

HARVARD MEDICAL SCHOOL

Pulseq For High-resolution Diffusion Acquisitions

Qiang Liu

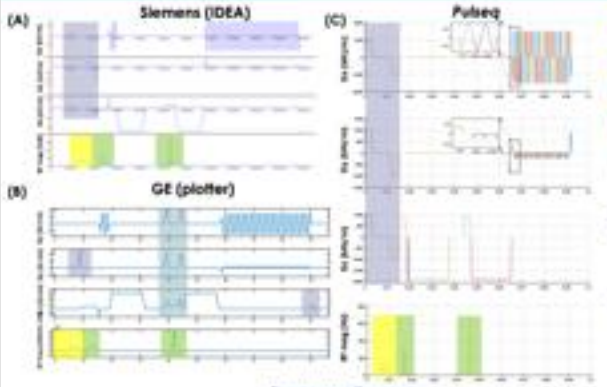
2026/02/26

Brigham and Women's Hospital, Harvard Medical School

Outline

Pulseq-diffusion sequence

https://github.com/HarmonizedMRI/Pulseq-diffusion



1. Implemented standard dMRI sequences using Pulseq
2. enhanced the **reproducibility** and consistency of dMRI measures across scanners
3. more than a **2.5×** reduction in variability across scanners (Phantom)

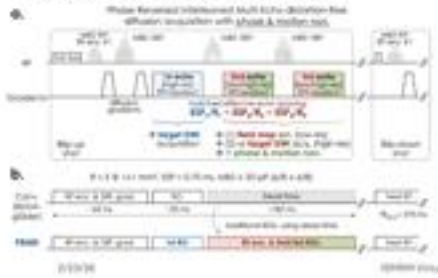
Pulseq-gSlider sequence for high resolution dMRI

- Use slab **RF encoding** and **thick-slice acquisition** to boost SNR
- Introduced to Pulseq with the function **makeArbitraryRf**

function [rf, gz, delay] = makeArbitraryRf(signal,flip,varargin)

- **PRIME:** Phase Reversed Interleaved Multi-Echo acquisition enables highly accelerated distortion-free diffusion MRI

gSlider RF pulses and slice profiles



Adding extra readouts without scan time penalty allows:

1. high-fidelity **field map** estimation from the 2nd echo
2. rapid SENSE-based multishot DWIs recon with **navigated** shot-to-shot phase correction
3. high-resolution **diffusion relaxometry** parameters were estimated (triple-echo readout)

https://arxiv.org/pdf/2409.07375
https://github.com/yohan-jun/PRIME

Pulseq gSlider with b-Tensor Encoding

- **Spherical Tensor Encoding (STE)** probes *microscopic diffusion anisotropy*
- Reveals microstructural features hidden to conventional **Linear Tensor Encoding (LTE)**
- Existing STE acquisitions suffer from **low spatial resolution**
- Increase spatial resolution of STE diffusion MRI without sacrificing SNR
- Key idea: STE encoding + gSlider slab RF encoding
- **mr.makeArbitraryGrad** with STE waveforms using **NOW** toolbox

Szczepankiewicz, F. et al. MRM 21'

ROVER-dMRI: Rotating-View Super-Resolution Reconstruction for High-SNR Mesoscale Diffusion MRI

1. Bridging macro-scale MRI (1–2 mm) and microstructure scale

Target resolution: **submillimeter / hundreds of microns**

Enables: Cortical layer-specific diffusion, fine white matter pathways, microstructural heterogeneity mapping

2. The **Challenge** of Going to Mesoscale: **SNR** drops rapidly with voxel size
3. Idea: acquire **high in-plane resolution,** use **thick slices** to boost SNR, rotate the FOV across scans

Result: high SNR, anisotropic resolution, rotated views provide complementary information

4. Classic super-resolution reconstruction (**SRR**) $\hat{x} = \arg\min \|Ax - y\|_2^2 + \lambda R(x)$
5. SRR with Implicit Neural Representation (**INR**)

$$M(\theta): \mathbb{R}^{dim} \to \mathbb{R}, x_i \mapsto M(\theta; x_i) = y_i$$

- x_i : spatial coordinate
- y_i : signal value at x_i
- M : neural network with parameters θ
- dim: spatial domain dimension

- INR represents images as **a continuous function**
- **Unsupervised**, without requiring paired high-resolution training data
- Implicit regularization effect, stabilize the ill-posed inverse problem

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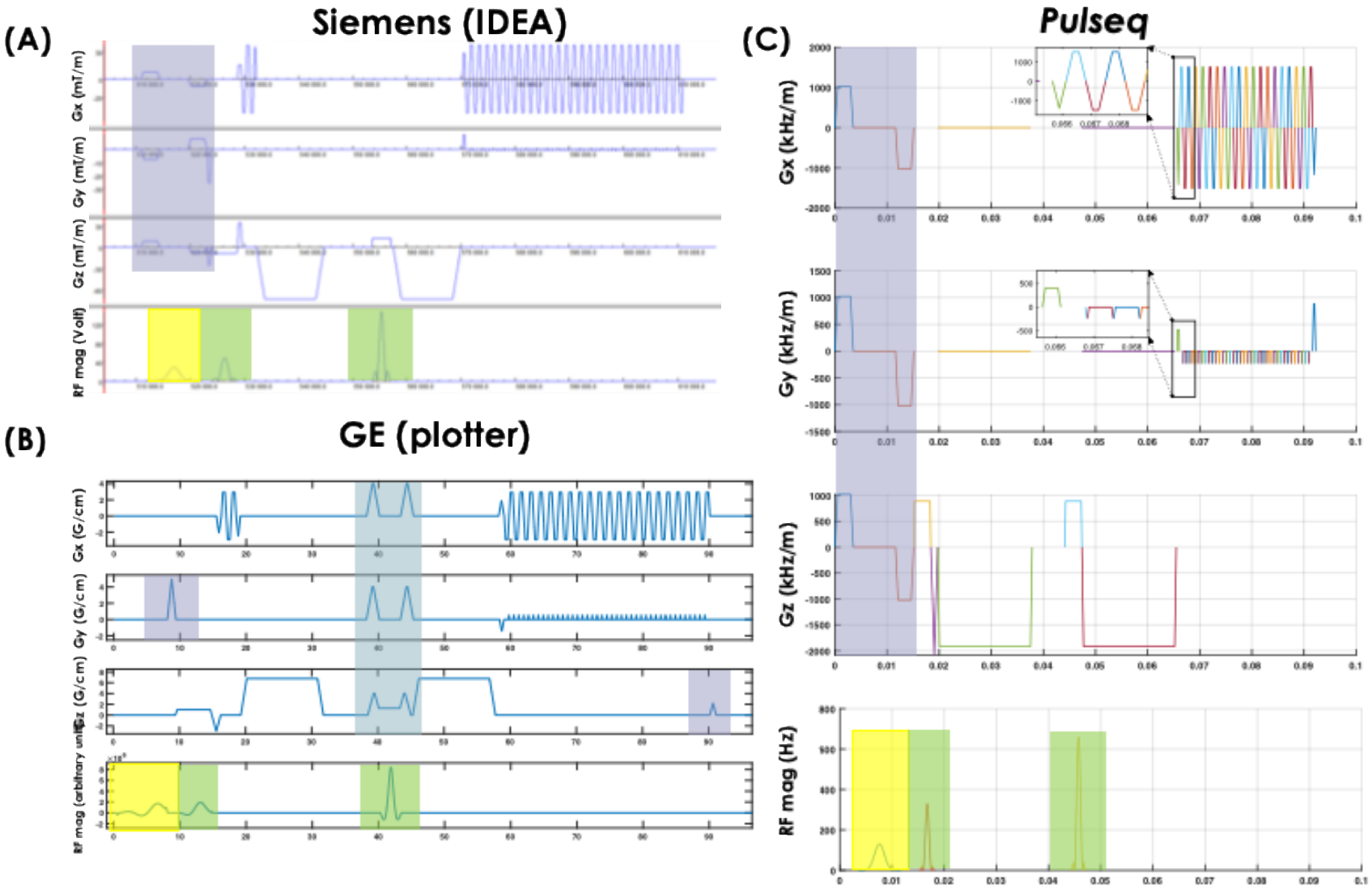

Sequence diagrams

- fat suppression
- ex & ref RFs
- spoiler
- crusher

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Liu Q et al. Reduced cross-scanner variability using vendor-agnostic sequences for single-shell diffusion MRI. Magn Reson Med. 2024;92(1):246-256.

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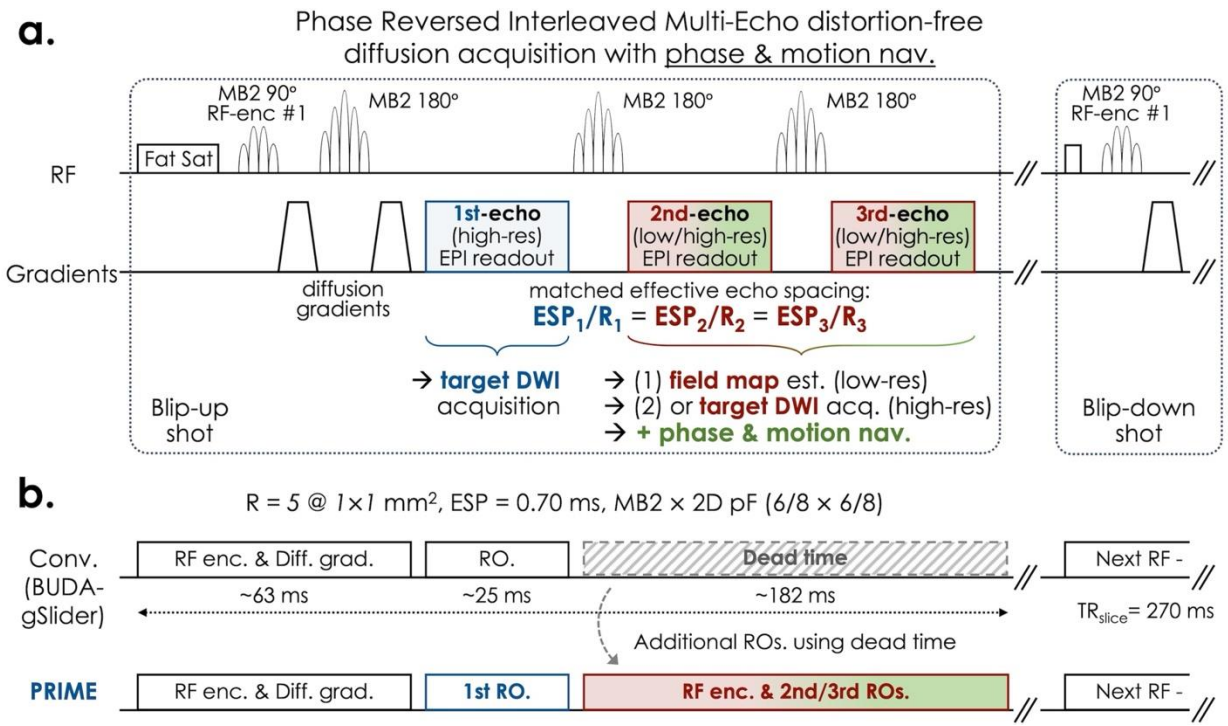

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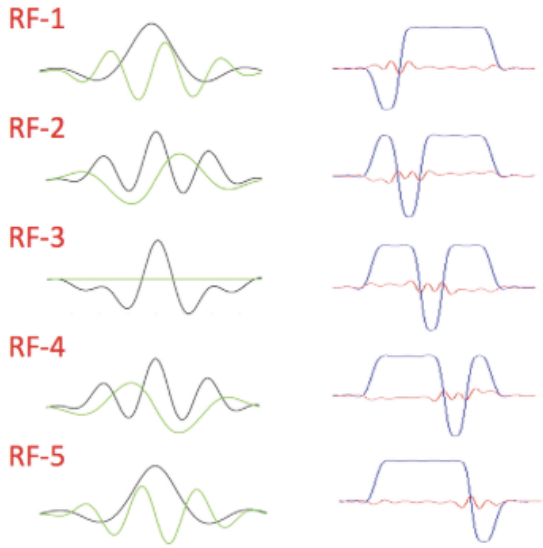

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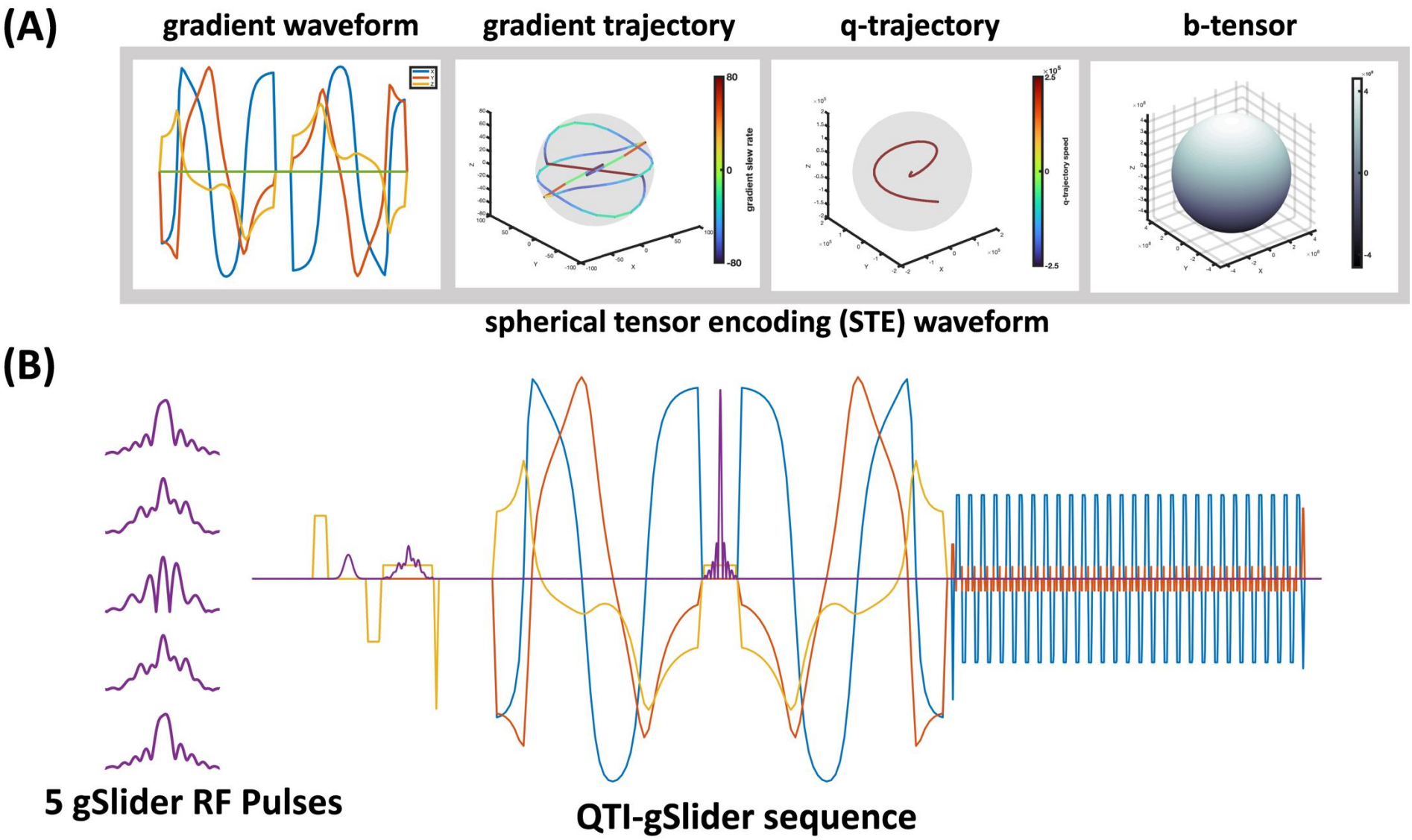

Szczepankiewicz, F. et al. MRM 21’

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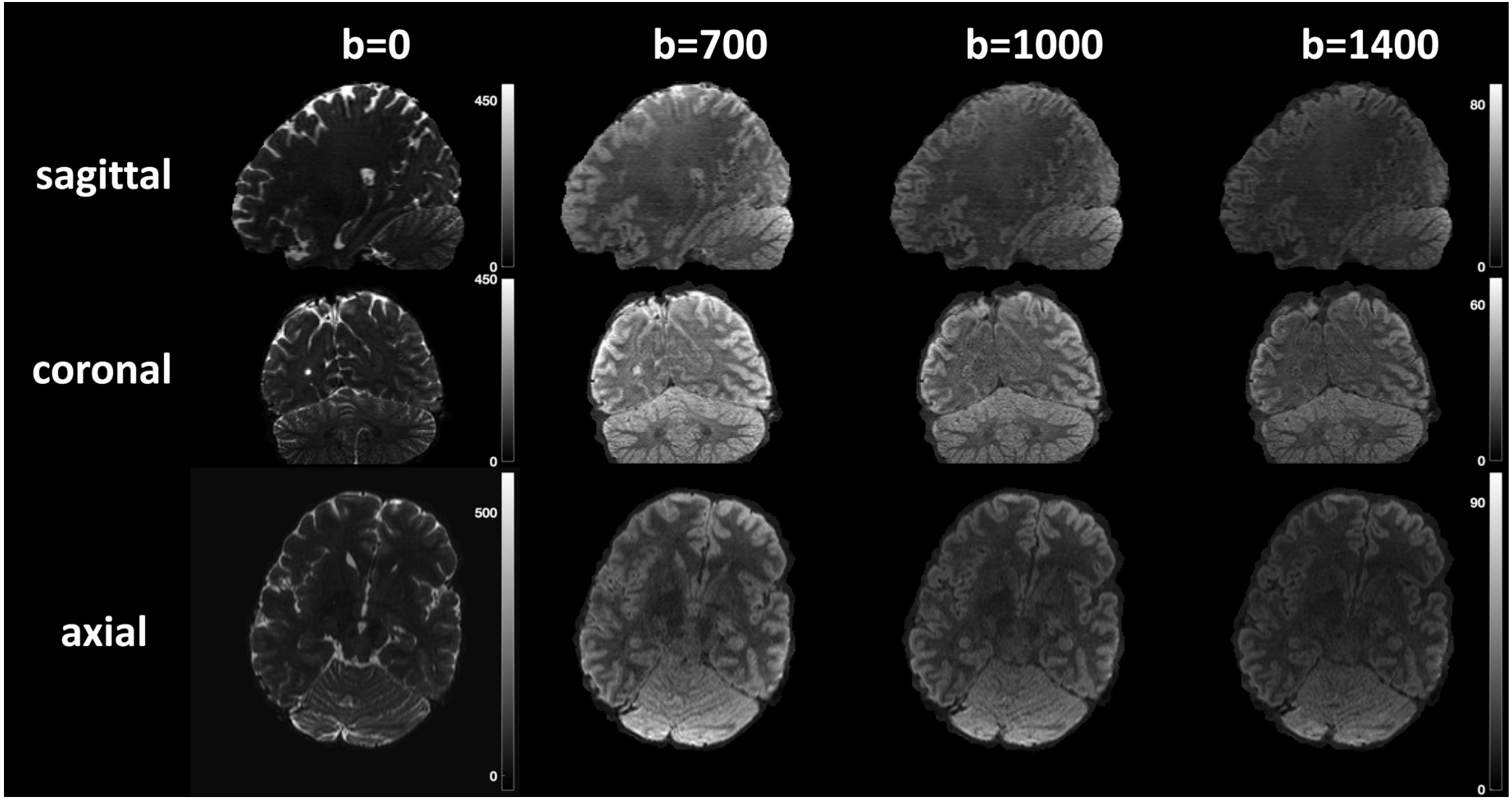

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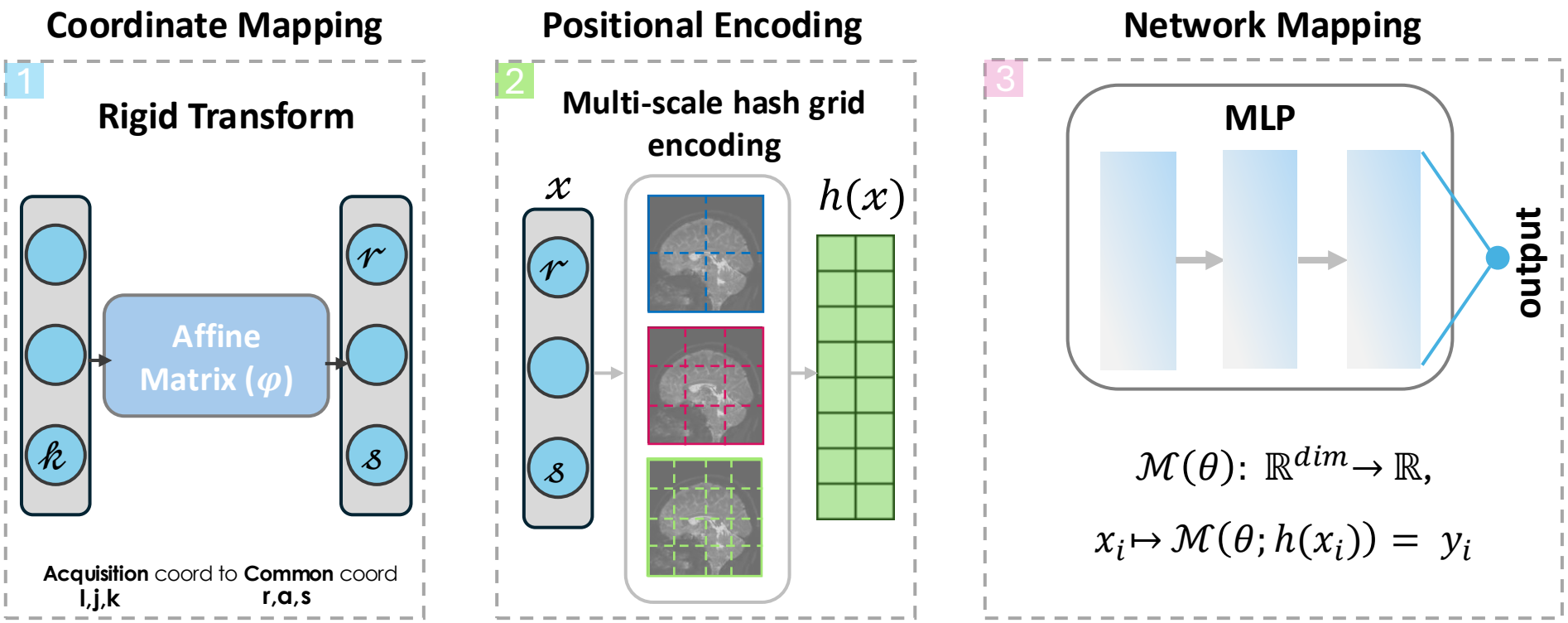
ROVER-dMRI framework

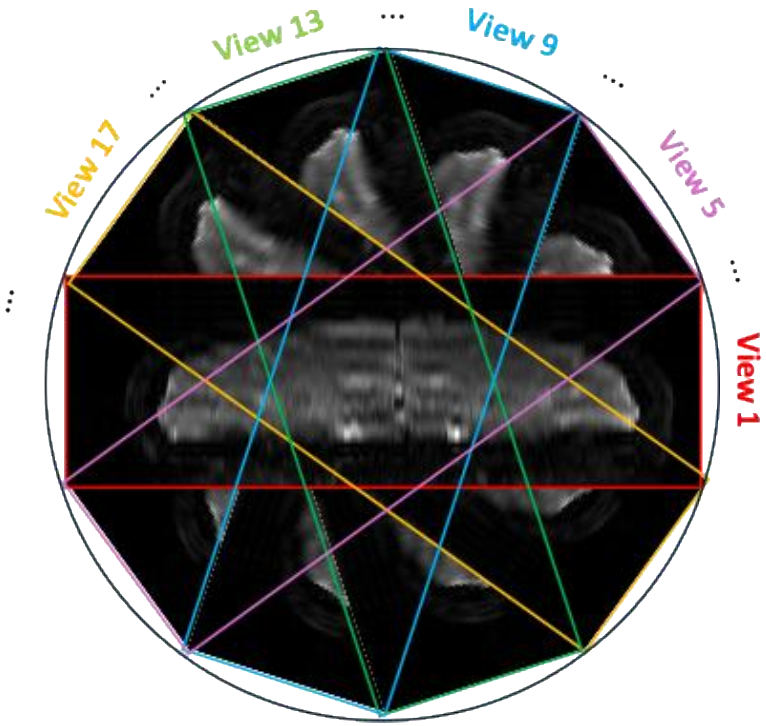


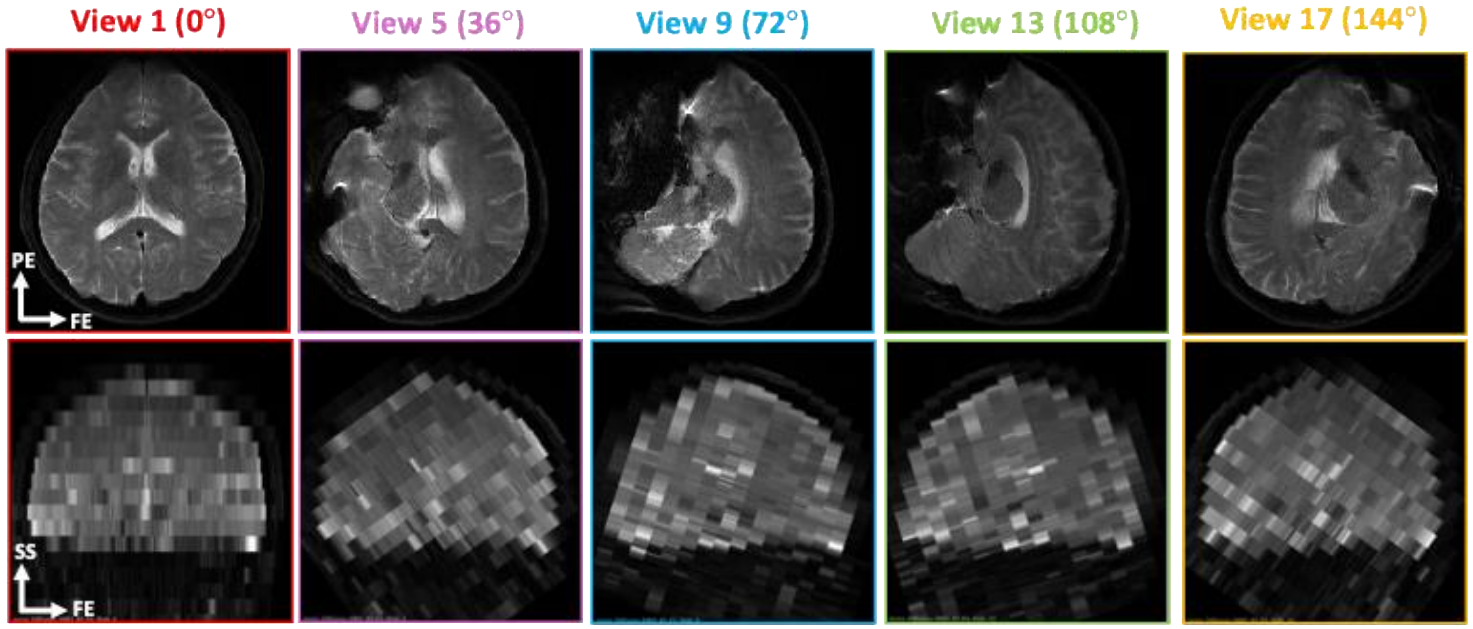
- Training used **mini-batch SGD** (batch size = 100k) with **data-fidelity** + **total-variation** losses.

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{N}\sum_{i=1}^{N}\left\| LR_i^v - \hat{y}_i^{\mathrm{avg}} \right\|_2^2 \qquad \mathcal{L}_{\mathrm{TV}} = \frac{1}{M}\sum_{j=1}^{M}\left\| \nabla \hat{y}_j \right\|_1 ,$$

Müller, T. et al., 2022.
Hendriks, et al., 2025.
Lyu, et al., 2025.

ROVER-dMRI image acquisitions – rotating views

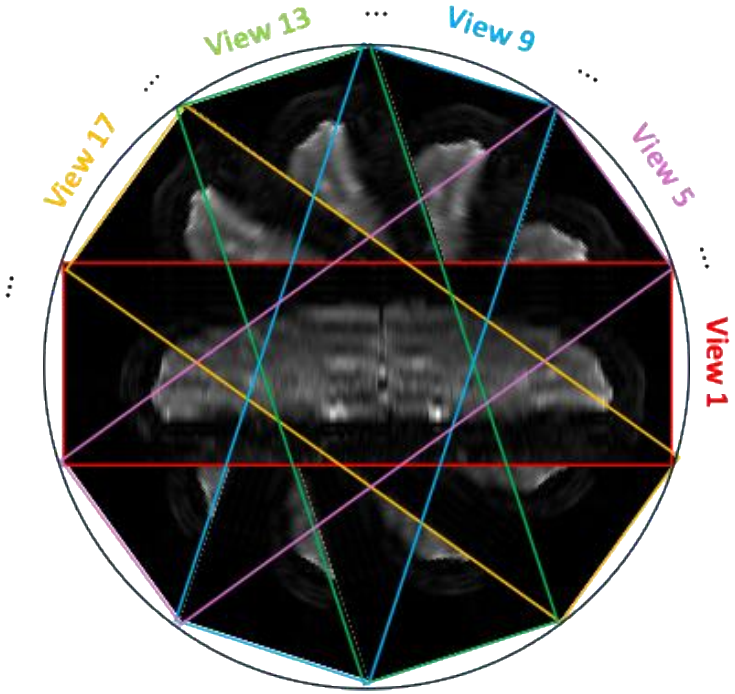


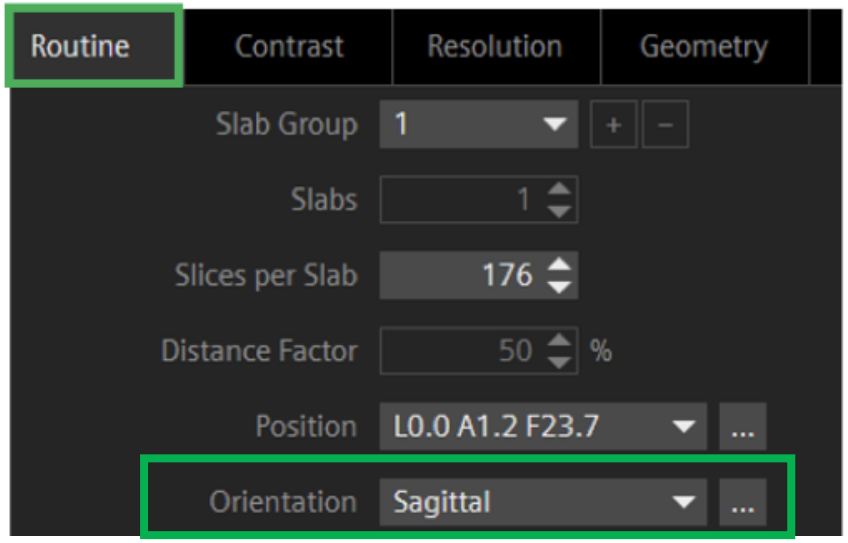


	FOV(mm^3)	Resolution (mm^3)	Multi-Band	In-plane Acc	Shot	TR/TE/ESP_{eff} (ms)	b-value(s/mm^2)	DWIs	Super-resolution factor	Rotation Views
Siemens Prisma	200×200×200	0.78×0.78×10	2	3	1	2000/80/0.35	1000	60	13	20
GE MAGNUS	220×220×168	0.44×0.44×6	1	3	2	3500/52.6/0.19	1000	25	13	20

ROVER-dMRI image acquisitions – rotating views

- **Rotate manually on the console (vendor-provided sequence)**





View #	Orientation
1	T 0
2	T-S 12
3	T-S 24
4	T-S 36
5	S-T 42
6	S-T 30
7	S-T 18

T : Transversal
S : Sagittal

ROVER-dMRI image acquisitions – rotating views with Pulseq

- **Rotate automatically in the Pulseq sequence (Pulseq V1.4.2)**

Rotation components

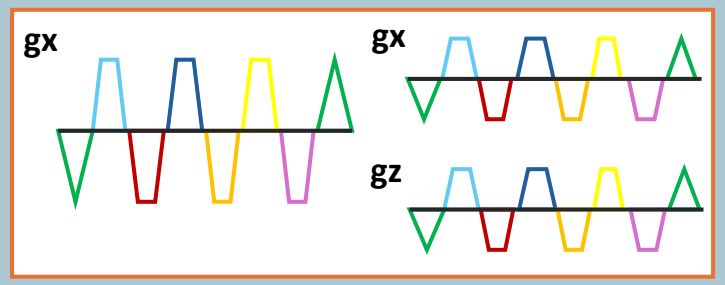
angle (φ)

axis (y)

Rotation matrix

$$\begin{bmatrix} \cos(\varphi) & 0 & \sin(\varphi) \\ 0 & 1 & 0 \\ -\sin(\varphi) & 0 & \cos(\varphi) \end{bmatrix}$$

Apply rotation to gradients



Transversal view Transverse-Sagittal 45°

mr.rotate/mr.rotate3D

```
%% sequence rotation
gx_comp1 = mr.rotate('y', alpha_rad, mr.scaleGrad(gx,dpg), mr.scaleGrad(gy_blipup, pe_enable*scale_gy_blipup),adc);
gx_comp2 = mr.rotate('y', alpha_rad, mr.scaleGrad(gx,dpg),mr.scaleGrad(gy_blipdown, pe_enable*scale_gy_blipdown),adc);
gx_comp3 = mr.rotate('y', alpha_rad, mr.scaleGrad(gx,dpg),mr.scaleGrad(gy_blipdownup, pe_enable*scale_gy_blipdownup),adc);

gxPre_post_comp = mr.rotate('y', alpha_rad, mr.scaleGrad(gxPre_post,dpg), mr.scaleGrad(gyPre_post,pe_enable*scale_gyPre_post));
gxPre_comp = mr.rotate('y', alpha_rad, mr.scaleGrad(gxPre,dpg),mr.scaleGrad(gyPre,pe_enable*scale_gyPre));

gz_comp = mr.rotate('y', alpha_rad, gz, rf);
gzReph_comp = mr.rotate('y', alpha_rad, gzReph);
gz180_comp = mr.rotate('y', alpha_rad, gz180, rf180);
```

ROVER-dMRI image acquisitions – rotating views with Pulseq

https://github.com/pulseq/pulseq

- **Rotate automatically in the Pulseq sequence (Pulseq V1.5.1)**

v1.5.1 Latest

m-a-x-i-m-z released this Aug 29, 2025 · 59 commits to master since this release v1.5.1 84ea726

A minor release with some important improvements and bug fixes. New features:

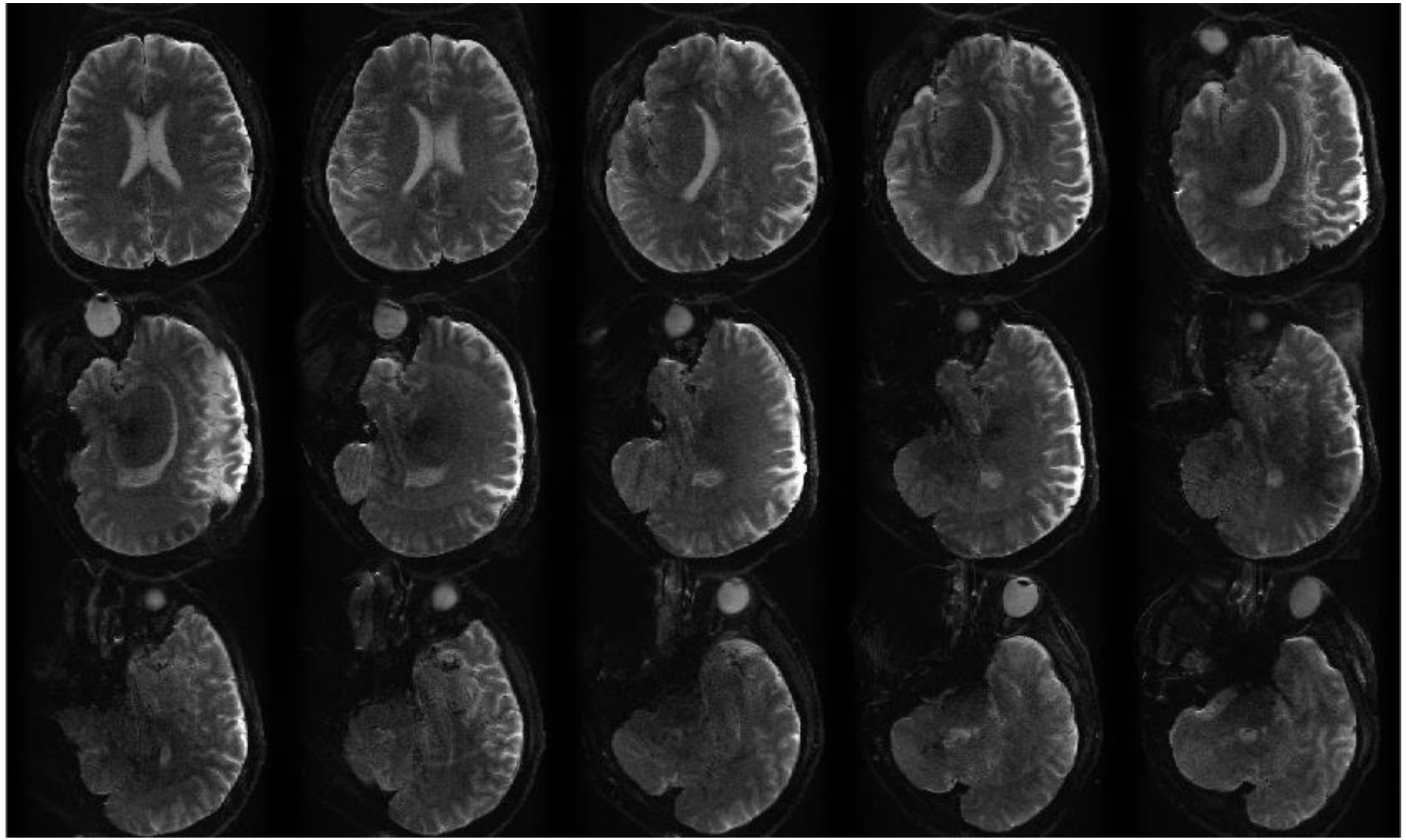
- Rotation extension
- RF shimming extension
- Plotting function shows triggers and digital output pulses
- Demo code for MR0 and Koma.jl signal simulations

```
for iview=[1:1]

    alpha_rad=rotation_angles(iview);
    rot_mat = [cos(alpha_rad), 0, sin(alpha_rad);0, 1, 0;-sin(alpha_rad), 0, cos(alpha_rad)];
    rot = mr.makeRotation(rot_mat);  % rotation event
                                ...
    seq.addBlock(gz,rf,rot,mr.makeLabel('SET', 'TRID', iview_fix));
    seq.addBlock(gzReph,rot);

                                ...
```

ROVER-dMRI – 0.4 mm iso with Pulseq on MAGNUS

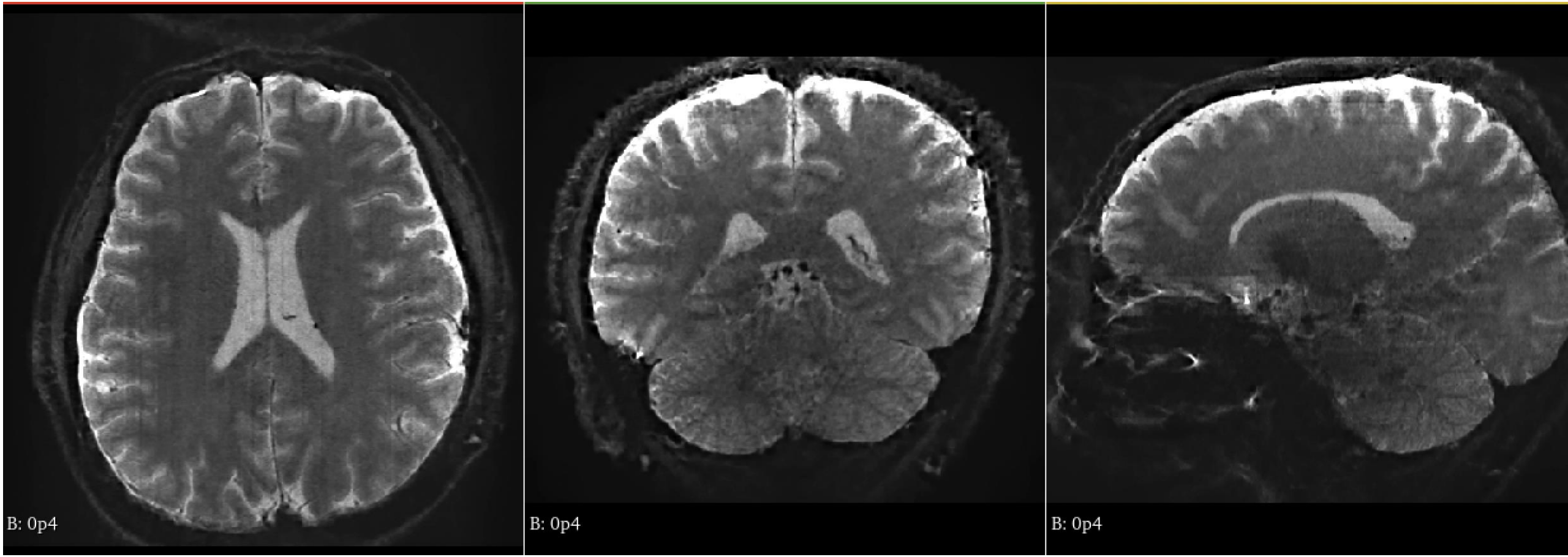
Low-resolution images

15 rotations, 0.4×0.4×6 mm^3, R=3



ROVER-dMRI – 0.4 mm iso with Pulseq on MAGNUS

Super-resolution images



0.4 mm iso

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Thank you for your attention!

qliu30@mgh.harvard.edu

Collaborators

Yunqi Wang
Nan Wang
Yiang Pan
Hwihun Jeong
Ante Zhu
Kawin Setsompop
Carl-Fredrik Westin
Berkin Bilgic
Borjan Gagoski
William Grissom
Lipeng Ning
Jon-Fredrik Nielsen
Maxim Zaitsev
Yogesh Rathi

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